



Prüfungsdeckblatt *Examination Cover Sheet*

Studienjahr <i>Academic Year</i>	2023/2024		
Studiengruppe(n) <i>Study Group(s)</i>	Resit		
Prüfung <i>Exam</i>	Klausur / <i>Written examination</i>		
Modul(e) <i>Module(s)</i>	Principles of Finance		
Zugelassene Hilfsmittel <i>Permitted aids</i>	Einsprachiges Wörterbuch, nicht-programmierbarer Taschenrechner / <i>monolingual dictionary, non-programmable calculator</i>		
Seiten Aufgabenstellung / Anlagen <i>Pages tasks / appendices</i>	18 Seiten inkl. Deckblatt / <i>18 pages incl. front page</i>		Formulär (4 pages)
Prüfungsdatum / -zeit / -dauer <i>Examination date/time/duration</i>	4 October 2024	8 – 9.30 a.m.	90 Minuten <i>Minutes</i>
Prüfer <i>Examiner</i>	Hr. Prof. Prigge (modulverantwortlich)		

Bitte leserlich ausfüllen! *Please complete legibly!*

Matrikelnummer <i>Matriculation number</i>			
Studiengruppe <i>Study Group</i>			
Anzahl zusätzlicher Seiten Papier <i>No. of additional sheets of paper</i>			
Klausurbeginn /-ende <i>Examination start / end</i>			

Punkteverteilung *Scores*

Aufgabe <i>Task</i>	1	2	3	4	Σ
Maximal <i>Maximum</i>	35	20	20	25	100
Erreicht <i>Achieved</i>					

Note Grade:	Prüfer Examiner:	Unterschrift Prüfer <i>Signature of the examiner:</i>

Punkte-Noten-Schema *Grading Scheme*

Punkte Score	100-95	94-90	89-85	84-80	79-75	74-70	69-65	64-60	59-55	54-50	49-0
Note Grade	1,0	1,3	1,7	2,0	2,3	2,7	3,0	3,3	3,7	4,0	n.b.

Task 1: Investment Decision Rules and WACC**(35 points)****a) Capital Budgeting**

The higher education department of the Super AI Tec AG considers realizing a project that would allow grading exams automatically with AI (called AI Grading). The project would require an initial investment outlay of EUR25 million and is expected to generate the future cash flows shown in EUR in the table below. The WACC for this project is 15%.

Year	1	2	3	4	5
Cash Flow	11.000.000	15.000.000	8.000.000	4.000.000	1.000.000

a1)

Calculate the NPV of the Project. Should Super AI Tec AG realize the project? **(12 points)**

	0	1	2	3	4	5
CF	-25.000.000	11.000.000	15.000.000	8.000.000	4.000.000	1.000.000
PV	-25.000.000,00	9.565.217,39	11.342.155,01	5.260.129,86	2.287.012,98	497.176,74
NPV	3.951.691,98					

NPV 10 points

Realize. 2 points

a2)

The higher education department is not the only department at Super AI Tec AG. The other departments also developed promising projects. However, as there is a shortage of AI engineers, it might be that not all projects with a positive NPV can be realized. The Chief Investment Officer of Super AI Tec AG compiled a list of all potential projects for next year, displaying their NPV and their need of AI engineer capacity (as headcount of AI engineers). What is the optimal investment program of Super AI Tec AG for next year, provided that they have a capacity of 80 AI engineers? Show your calculations. **(9 points)**

Project	NPV (EUR)	Required AI Engineer Capacity
AI Grading	result from a1)	20
P1	2.500.000,00	15
P2	3.500.000,00	18
P3	6.000.000,00	25
P4	6.500.000,00	35
P5	7.800.000,00	43
P6	3.300.000,00	17

Project	NPV (EUR)	Required AI Engineer Capacity	PI	Rank	Capacity Used
AI Grading	3.951.691,98	20	197.584,60	2	45
P1	2.500.000,00	15	166.666,67	7	
P2	3.500.000,00	18	194.444,44	3	63
P3	6.000.000,00	25	240.000,00	1	25
P4	6.500.000,00	35	185.714,29	5	
P5	7.800.000,00	43	181.395,35	6	
P6	3.300.000,00	17	194.117,65	4	80

1 point for each PI → **7 points**

The firm should realize the project with the highest PIs unless the engineer capacity is exhausted. In this case, AI Grading and the projects P2, P3, and P6 should be realized. **2 points**

b) WACC

For Super AI Tec AG, the following data is given:

- cost of equity 17%
- cost of debt 10%
- market value of equity EUR250 million
- market value of debt EUR200 million
- corporate tax rate 35%

b1) Please calculate Super AI Tec AG's cost of debt after taxes. **(3 points)**

$$(1-0,35) * 10\% = 6.5\%$$

b2) Please calculate Super AI Tec AG's company WACC. **(6 points)**

$$WACC = 200/(200+250)*(1-0.35)*10\% + 250/(200+250)*17\% = 12.33\%$$

b3) The company WACC you just calculated differs from the WACC applied in task a1). Why might this make sense? How can the difference be interpreted? **(5 points)**

Makes sense that project WACC differs from company WACC when risk profile of the project differs from that of the company. **2.5 points**

In this case, as the project WACC is higher than the company WACC, it seems that the project is riskier than the current business activity of the company. **2.5 points**

Aufgabe Task 2 Bonds**(20 Punkte Points)**

Suppose that today you buy an 8% annual coupon bond for EUR1,150. The bond has 10 years to maturity and a face value of EUR1,000.

- a) If you hold the bond until maturity, what rate of return do you expect to earn on your investment (no approximation via interpolation required)? **(4 points)**

The YTM for holding it until maturity is:

$$CPN = \frac{\text{Coupon Rate} \cdot \text{Face Value}}{\text{number of Coupon Payments per Year}} = \frac{8\% \cdot 1000}{1} = 80$$

$$1,150 = 80 \times \frac{1}{y} \left(1 - \frac{1}{(1+y)^{10}} \right) + \frac{1,000}{(1+y)^{10}}$$

$$\text{YTM} = 5.97\%$$

a)				
current price	1150			
face value	1000			
coupon rate	0,08			
n	10			
i(market)	5,97%			
Year	Cash flow	q	q^n	PV
0	-1150			
1	80	1,05965327	1,05965327	75,4963931
2	80	1,05965327	1,12286506	71,2463172
3	80	1,05965327	1,18984763	67,2354997
4	80	1,05965327	1,26082594	63,4504715
5	80	1,05965327	1,33603833	59,8785217
6	80	1,05965327	1,41573738	56,5076552
7	80	1,05965327	1,50019075	53,3265519
8	80	1,05965327	1,58968204	50,3245291
9	80	1,05965327	1,68451177	47,4915054
10	1080	1,05965327	1,78499841	605,042555
			PV	1150

- b) Two years from now, the yield to maturity (YTM) on your bond has declined by one percentage point and you decide to sell. What price will your bond sell for? [Hint: If you did not calculate the initial YTM under a), assume the initial YTM two years ago was 7%.] **(4 points)**

The price of the bond in two years, at the new YTM, will be:

$$P = 80 \times \frac{1}{0.0497} \left(1 - \frac{1}{(1.0497)^8} \right) + \frac{1000}{(1.0497)^8} = \mathbf{1,196.41}$$

b)				
face value	1000			
coupon rate	0,08			
n	8			
i(market)	4,97%			
Year	Cash flow	q	q^n	PV
1	80	1,04965327	1,04965327	76,2156439
2	80	1,04965327	1,10177199	72,6103047
3	80	1,04965327	1,15647858	69,1755141
4	80	1,04965327	1,21390152	65,9032044
5	80	1,04965327	1,2741757	62,7856895
6	80	1,04965327	1,33744269	59,8156469
7	80	1,04965327	1,4038511	56,9861006
8	1080	1,04965327	1,4735569	732,920459
			PV	1.196,41

- c) Assuming you sold the bond at the price you calculated under b), what is the internal rate of return on your investment over your holding period of two years? Approximation via linear approximation is required. Please calculate two NPVs, one positive and one negative, using the interest rates of 8% and 9%. **(6 points)**

$$1,150 = 80 \times \frac{1}{y} \left(1 - \frac{1}{(1+y)^2} \right) + \frac{1,196.46}{(1+y)^2}$$

$$NPV = -1,150 + \frac{80}{y} \left(1 - \frac{1}{(1+y)^2} \right) + \frac{1,196.46}{(1+y)^2}$$

$$NPV_1(8\%) = 18.39; NPV_2(9\%) = -2.27$$

$$r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1) = \mathbf{8.89\%}$$

c)							
buying price	1150						
selling price	1.196,41						
face value	1000						
coupon rate	0,08						
n	2						
i(1)	0,08						
i(2)	0,09						
		NPV(1)			NPV(2)		
Year	Cash flow	q	q^n	PV	q	q^n	PV
0	-1150			-1150			-1150
1	80	1,08	1,08	74,0740741	1,09	1,09	73,3944954
2	1276,41256	1,08	1,1664	1094,31804	1,09	1,1881	1074,33092
			NPV	18,3921151		NPV	-2,2745871
	IRR interpoliert		8,89%				
	IRR Excel		8,89%				

- d) Compare this return c) to the rate of return a) when you first bought the bond. Why are they different? [If you did not calculate a return under c) and not calculated one under a), then use a return for c) of 10% and a return for a) of 7%.] **(2 points)**

The realized internal rate of return is greater than the expected YTM when the bond was bought because YTM dropped by one percentage point; bond prices rise when yields fall. [same applies for returns for c) of 10% and a return for a) of 7%]

- e) Assuming after three years, the price of the bond will be at EUR900. Would the bond be sold at a premium or a discount? Has the market interest rate fallen or risen over the three years? Explain briefly. **(2 points)**

The bond would be sold at a discount. Discount bond because price will be below face value. Market interest rate will rise.

- f) If an investor holds a bond trading at a discount, what sources of return does he expect? **(2 points)**

If a coupon bond trades at a discount, an investor will earn a return both from receiving the coupons and from receiving a face value that exceeds the price paid for the bond.

a) NewVentures AG expects an earnings per share (EPS) of EUR5.20 for the running business year. As the management plans to pursue a promising investment opportunity with 17% expected rate of return in the following year, they will retain 80% of this year's EPS. For all the years thereafter, they only have fewer promising investments with an expected return of 12%. Therefore, they will only retain 50% of earnings for those following years. Assume that any leftovers are paid out as dividends to shareholders. Please calculate NewVentures' current stock price at equity cost of capital of 8%. **(14 points)**

$$g_2 = 0.8 \times 0.17 = 0.136 \text{ (2 points)}$$

$$\text{EPS}_2 = 5.20 \times 1.136 = 5.91 \text{ (2 points)}$$

$$\text{DIV}_1 = 0.2 \times 5.20 = 1.04 \text{ (1 point)}$$

$$\text{DIV}_2 = 0.5 \times 5.91 = 2.96 \text{ (1 point)}$$

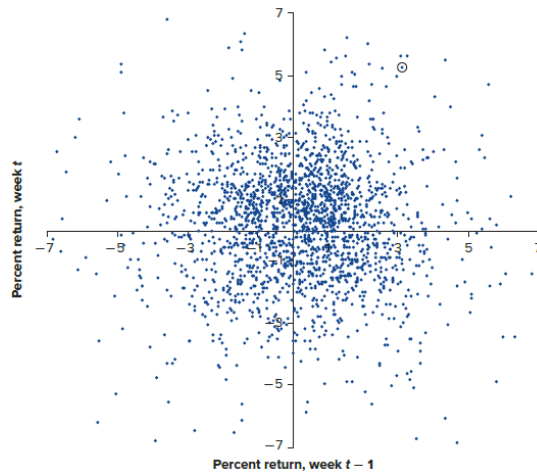
$$g_\infty = 0.5 \times 0.12 = 0.06 \text{ (2 points)}$$

$$P_0 = \frac{1.04}{1.08} + \frac{\left(\frac{2.96}{0.08-0.06}\right)}{1.08} = 138.00 \text{ (6 points)}$$

(solution if $\text{EPS}_2 / \text{DIV}_2$ are rounded, exact solution see below)

t		1	2	3
EPS		5,20	5,91	6,26
g		-	0,136	0,06
Return on new investment		-	0,17	0,12
Retention rate		0,8	0,5	0,5
Dividend payout rate		0,2	0,5	0,5
DIV		1,04	2,95	3,13
Perpetuity			0,02	
Discount rate		1,08		
PV	137,70	0,962963	136,7407	

b) Please explain briefly the depicted chart below. Which form of market efficiency can be described with it? In how far can you outperform the market in that form of efficiency? **(6 points)**



The chart describes the “random walk theory” showing that there is no correlation between stock returns between two successive weeks. This means that future stock prices cannot be predicted based on historic market information as all historic information is already incorporated into those prices. This defines the weak form of market efficiency. However, in weak form efficient markets investors might still be able to outperform the market by analyzing publicly available key figures and other specific data of the respective company to uncover under- or overvalued stocks.

- a) Please explain the differences and commonalities between the risk profile of a single stock with a $\beta=1$ and a perfectly diversified portfolio with a $\beta=1$. **(5 points)**

Both positions have the same sensitivity with regard to systematic risk. However, total risk of the portfolio includes only systematic risk, whereas total risk of the single stock additionally includes specific risk. Thus, total risk of the single stock is bigger than that of the portfolio.

- b) You already hold Stock 1 in your stock portfolio. Inspired by your Finance lectures, you would like to start to diversify your stock portfolio. You like to add either Stock 2 or Stock 3 to your portfolio. After this transaction, both stocks in your portfolio have a weight of 50%. Assume that all economic scenarios are equally likely.

	Recession	Normal	Boom	Expected Return	Standard Deviation
Stock 1	-9.00%	8.00%	21.00%	6.67%	12.28%
Stock 2	-0.07%	5.00%	15.00%	4.33%	8.99%
Stock 3	10.00%	4.00%	-5.00%	3.00%	6.16%

- b1) Looking at the data, would you expect a bigger diversification effect from adding Stock 2 or Stock 3 to your portfolio? Explain. **(5 points)**

Stock 3 because it is countercyclical, whereas Stock 2 is like Stock 1 procyclical. Thus, correlation between Stock 1 and Stock 3 will be much lower than between Stock 1 and Stock 2.

- b2) Please calculate the expected return of the portfolio after the second stock has been added. **(4 points)**

- b3) Please calculate the standard deviation of the portfolio after the second stock has been added. **(6 points)**

	Recession	Normal	Boom	Expected Return	Standard Deviation
Stock 1	-9,00%	8,00%	21,00%	6,67%	12,28%
Stock 3	10,00%	4,00%	-5,00%	3,00%	6,16%
Portfolio	0,50%	6,00%	8,00%	4,83%	3,17%

- c) Applying the CAPM, what is the expected return for a stock with a beta=1.2 when the risk-free rate is at 3% and the market risk premium amounts to 7%? **(5 points)**

$$r(E) = 3\% + 1.2 * 7\% = 11.4\%$$

Formulary – Appendix to Exam

Chapter 1

Future Value of a Cash Flow: $FV_n = C \times (1 + r) \times (1 + r) \times \dots \times (1 + r) = C \times (1 + r)^n$

Present Value of a Cash Flow: $PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$

Net Present Value (NPV): compares the present value of cash inflows (benefits) to the present value of cash outflows (costs).

Present Value of a Perpetuity (first payment at time 1): $PV(C \text{ in perpetuity}) = \frac{C}{r}$

Present Value of an Annuity: $PV(\text{annuity of } C \text{ for } N \text{ periods with interest rate } r) = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right)$

Future Value of an Annuity:

$$\begin{aligned} FV(\text{annuity}) &= PV \times (1 + r)^N \\ &= \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right) \times (1 + r)^N \\ &= C \times \frac{1}{r} ((1 + r)^N - 1) \end{aligned}$$

Present Value of a Growing Perpetuity: $PV(\text{growing perpetuity}) = \frac{C}{r - g}$

Present Value of a Growing Annuity: $PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{(1 + r)}\right)^N\right)$

Approximate Internal Rate of Return (IRR) via Interpolation
(with $NPV_1 > 0$ and $NPV_2 < 0$): $r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$

Chapter 2

Profitability Index: $\text{Profitability Index} = \frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{NPV}{\text{Resource Consumed}}$

Chapter 3

Coupon Payment: $CPN = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of Coupon Payments per Year}}$

Price of a Zero-Coupon Bond (FV=face value): $P = \frac{FV}{(1 + YTM)^n}$

Yield to Maturity of an n-Year Zero-Coupon Bond: $YTM = \left(\frac{FV}{P}\right)^{\frac{1}{n}} - 1$

Yield to Maturity of a Coupon Bond: discount rate y , that equates the PV of the bond's future cash flows with its price (=IRR): $P = CPN \times \frac{1}{YTM} \left(1 - \frac{1}{(1 + YTM)^N}\right) + \frac{FV}{(1 + YTM)^N}$

Chapter 4

Dividend-discount model (1 year): $P_0 = \left(\frac{\text{Div}_1 + P_1}{1 + r_E} \right)$

Total return, capital gain, dividend yield: $r_E = \frac{\text{Div}_1 + P_1}{P_0} - 1 = \underbrace{\frac{\text{Div}_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$

Dividend-discount model (2 years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2 + P_2}{(1 + r_E)^2}$

Dividend-discount model (n years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$

$$P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \frac{\text{Div}_3}{(1 + r_E)^3} + \dots = \sum_{n=1}^{\infty} \frac{\text{Div}_n}{(1 + r_E)^n}$$

DDM, constant growth: $P_0 = \frac{\text{Div}_1}{r_E - g} \quad r_E = \frac{\text{Div}_1}{P_0} + g$

DDM, changing growth rates: $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N} \left(\frac{\text{Div}_{N+1}}{r_E - g} \right)$

Change in Earnings = New Investment × Return on New Investment

New Investment = Earnings × Retention Rate

Earnings Growth Rate = $\frac{\text{Change in Earnings}}{\text{Earnings}} = \text{Retention Rate} \times \text{Return on New Investment}$

$g = \text{Retention Rate} \times \text{Return on New Investment}$

Dividend: $\text{Div}_t = \frac{\text{Earnings}_t}{\underbrace{\text{Shares Outstanding}_t}_{\text{EPS}_t}} \times \text{Dividend Payout Rate}_t$

Chapter 5

Expected (Mean) Return: Calculated as a weighted average of the possible returns, where the weights correspond to the probabilities.

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

Variance: The expected squared deviation from the mean.

$$\text{Var}(R) = E[(R - E[R])^2] = \sum_R P_R \times (R - E[R])^2$$

Variance Estimate Using Realized Returns:

$$\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$$

Standard Deviation: The square root of the variance.

$$SD(R) = \sqrt{\text{Var}(R)}$$

$$\begin{aligned} \text{Portfolio rate of return} &= (\text{fraction of portfolio in asset 1} \times \text{rate of return on asset 1}) \\ &+ (\text{fraction of portfolio in asset 2} \times \text{rate of return on asset 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in asset } t \times \text{rate of return on asset } t) \end{aligned}$$

$$\begin{aligned} \text{Portfolio beta} &= (\text{fraction of portfolio in stock 1} \times \beta \text{ of stock 1}) \\ &+ (\text{fraction of portfolio in stock 2} \times \beta \text{ of stock 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in stock } t \times \beta \text{ of stock } t) \end{aligned}$$

$$\text{Risk premium on the market} = r_m - r_f$$

$$\text{Risk premium on any asset } i = r_i - r_f$$

$$\text{SML equation: } r = r_f + \beta(r_m - r_f)$$

Chapter 6

$$r_{\text{assets}} = \frac{\text{total income}}{\text{value of investments}} = \frac{(D \times r_{\text{debt}}) + (E \times r_{\text{equity}})}{V}$$

$$r_{\text{assets}} = \frac{D}{V} \times r_{\text{debt}} + \frac{E}{V} \times r_{\text{equity}}$$

$$\text{After-tax cost of debt} = r_{\text{debt}} \times (1 - T_c).$$

$$\text{WACC} = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left[\frac{E}{V} \times r_{\text{equity}} \right]$$